The First Law of Thermodynamics The Microscopic Model As you can see here, we are moving

[H]
[width=0.6] /PhyAI/chapter1/$1_0.png$ Our hypothesis model of gas and crystal But why do I call these particles "imaginary
[width=0.7] /PhyAI/chapter1/$1_1.png$ Definition of Volume Now you might wonder if you mix 2 types of completely differen
[width=0.7] /PhyAI/chapter1/$1_2.png$ Mixing $H2$ and He together Reaching the equilibrium state means the density of gas

Unlike volume, **pressure** is very easy concept to define: [The First Definition of Pressure] Pressure is the force applied

^{2}), or (Pa)

One classic example of pressure is pressing your hand against a small piston filled with gas. If you press hard and fast enou
[width=0.6] /PhyAI/chapter1/$1_3.png$ Pressing a small piston After understanding what volume and pressure are, now it is
[width=0.8] /PhyAI/chapter1/$1_4.png$ For every degree decrease, the volume $V/273$ decreases correspondingly But because

This section does not have much to discuss beyond introducing an important law of gases and some new constants. As you
[width=0.5] /PhyAI/chapter1/$1_5.png$ An oversimplified model of an ideal gas Since the system of 10^{23} particles is impossib
My oversimplified model of an ideal gas contains only 1 particle.
Its speed never changes.
Piston is perfectly smooth, so collisions are always **elastic**.
Directly applying Definition yields: $\overline{P}_{x,onpiston} = \frac{\overline{F}_{x,onpiston}}{A} = -\frac{\overline{F}_{x,onparticle}}{A}(Newton'sThirdLaw)$
$= -\frac{m\dfrac{\Delta v}{\Delta t}}{A} = -\frac{m\dfrac{(-v_x) - v_x}{2L/v_x}}{A} = \frac{mv_x^2}{V}$
If our piston contains N particles, then the total pressure on piston by x-axis is:

Because N is extremely large, and the gas is homogeneous:

The average velocity of the particles in the x, y and z directions is the same, we obtain:

The average velocity of N particles in three-dimensional space is defined as:

The average translational kinetic energy is then:

We can simply explain the phenomenon of thermal expansion using the previous result. If the temperature increases, the av
Moving along x-axis: $\overline{K}_x = \frac{1}{2}m\overline{v_x^2} = \frac{k_BT}{2}$
Moving along y-axis: $\overline{K}_y = \frac{1}{2}m\overline{v_y^2} = \frac{k_BT}{2}$
Moving along z-axis: $\overline{K}_z = \frac{1}{2}m\overline{v_z^2} = \frac{k_BT}{2}$
Other degrees of freedom include rotational motion and elastic potential energy (as stored in a structure like a spring), and

The final problem is determining the value of f. For example, if our gas is monatomic, then we can easily conclude that f =
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Rotating around x-axis: $\overline{K}_{x,rotate} = \frac{1}{2}I\omega_x^2 = \frac{k_BT}{2}$
Rotating around y-axis: $\overline{K}_{y,rotate} = \frac{1}{2}I\omega_y^2 = \frac{k_BT}{2}$
Most polyatomic molecules can rotate about all three axes, so if our gas is polyatomic, then $f = 6$. In the case of solid, f =
[width=0.8] /PhyAI/chapter1/$1_6.png$ Degrees of freedom illustration Counting degrees of freedom of liquid is more complic

Applying Corollary , we obtain: